

TABLE VI: Characteristics for Assessing Fuzz Driver Quality

| ID  | Element                 | Description (S=2 Good, S=1 Fair, S=0 Poor)                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1  | API Usage               | How well the fuzz driver correctly invokes the target API functionality.<br>(S=2) Fully implements the relevant API behavior.<br>(S=1) Implements only part of the relevant API behavior.<br>(S=0) Fails to invoke the API properly, causing validation errors or false crashes.                                                                                                                                                            |
| Q2  | Cleanup                 | How well allocated resources are released through appropriate cleanup and free calls.<br>(S=2) All allocations are cleaned up.<br>(S=1) Only some allocations are cleaned up.<br>(S=0) No cleanup is performed.                                                                                                                                                                                                                             |
| Q3  | Dynamic Parameters      | Whether the fuzz driver varies important parameters rather than fixing them to constant values.<br>(S=2) All important parameters are dynamic.<br>(S=1) Some parameters are dynamic and others are hardcoded.<br>(S=0) All important parameters are hardcoded.                                                                                                                                                                              |
| Q4  | Error Cases             | Whether the fuzz driver intentionally exercises error-handling behavior by supplying inputs that should trigger failure paths.<br>(S=2) All identifiable error-handling cases are exercised.<br>(S=1) Some error-handling cases are exercised.<br>(S=0) No error-handling cases are exercised.                                                                                                                                              |
| Q5  | Rare Branches           | Whether the fuzz driver reaches specialized or rarely exercised branches of the target code.<br>(S=2) All identified rare branches are exercised.<br>(S=1) Some rare branches are exercised.<br>(S=0) No rare branches are exercised.                                                                                                                                                                                                       |
| Q6  | Realistic Data          | How well the fuzz driver reconstructs the data structures and parameter dependencies normally expected by the target API.<br>(S=2) Data structures and parameter ranges are reproduced accurately and exercised broadly.<br>(S=1) Data structures are simplified or somewhat inaccurate, or parameter ranges are overly constrained.<br>(S=0) Data is passed only in trivial ways or with invalid values that do not reflect realistic use. |
| Q7  | Sequences of Functions  | The effectiveness of the sequence of function calls made by the fuzz driver.<br>(S=2) Function calls are sequenced meaningfully, with later calls building on earlier ones to exercise deeper behavior.<br>(S=1) The sequence is logical but limited in its contribution to coverage.<br>(S=0) Calls are effectively independent or ordered without meaningful dependence.                                                                  |
| Q8  | Function Call Variation | Whether the fuzz driver varies its call sequence across executions.<br>(S=2) The driver meaningfully varies call sequences, for example through conditional or repeated behavior.<br>(S=1) Some variation occurs through limited conditional logic.<br>(S=0) Functions are called in exactly the same order every time.                                                                                                                     |
| Q9  | Validity of Inputs      | How consistently the fuzz driver exercises the target under valid or semantically plausible inputs.<br>(S=2) The driver avoids false crashes and may include checks for unstable or flaky behavior.<br>(S=1) The driver usually behaves plausibly but occasionally exhibits flaky or weakly constrained behavior.<br>(S=0) The driver frequently causes invalid executions due to bad call sequences or parameters.                         |
| Q10 | Verification of Output  | How thoroughly the fuzz driver examines the result of function calls.<br>(S=2) Output is explicitly verified, for example through assertions or round-trip checks.<br>(S=1) Success or failure is checked for some but not all function calls.<br>(S=0) Output is discarded without examination.                                                                                                                                            |